

Airy logarithmic asymptotics for bounded indegree labelled DAGs

Exact parking reduction and a Lambert W inverse

Report 157

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Abstract

Let a_n count simple directed acyclic graphs on the labelled vertex set $[n]$ with every indegree at most two, the sequence OEIS A397711. If $-\zeta$ is the largest real zero of the Airy function, then

$$\log a_n = 2n \log n - n - \kappa n^{1/3} + o(n^{1/3}), \quad \kappa = \frac{3\zeta}{2^{1/3}}.$$

An exact reduction through generalized parking functions represents the count as a Poisson excursion partition function with potential $q_i = \log(i/(i-1)) \sim i^{-1}$. Geometric blocks transfer established homogeneous area-tilt bounds of Mallein to this singular potential. Explicit interface intervals, a fixed initial segment, and a forced terminal descent handle the boundaries and the growing number of blocks.

For $y = \log x$ and $b = y/[2W_0(y/(2\sqrt{e}))]$, the threshold inverse is

$$\min\{n \geq 1 : a_n \geq x\} = b + \frac{\kappa b^{1/3}}{2 \log b + 1} + o\left(\frac{b^{1/3}}{\log b}\right).$$

The same argument gives the logarithmic Airy correction for every fixed integer indegree bound $c \geq 2$. Exact enumeration, generalized parking functions, and the homogeneous Airy estimates are prior inputs. The result is a logarithmic asymptotic; no power prefactor, amplitude, $O(\log n)$ remainder, or all-orders expansion is established.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 157 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a397711-bounded-indegree-dags** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 157” and its PDF author field is empty. *What rests on what.* The sink recurrence (Proposition 2.1), the monotonicity bound (7.1), the occupancy count (2.8) from (2.5), and the exact normalization (Proposition 3.1) are proved here; the step to (2.5) uses the general-vector parking identity of Kung and Yan [4], cited with the source’s sketch. The analytic core, Lemma 5.1, is proved here *from* the two homogeneous killed-walk estimates of Proposition 4.1, which are Mallein’s Lemmas 2.6 and 2.7 [6]: an external input, cited and not reproved, whose statements the write read (note in Section 4); nobody in this intake verified Mallein’s proofs. Theorems 1.1 and 1.3 follow from the lemma, the exact normalization and Stirling’s formula, and Theorem 1.2 from Theorem 1.1 and (7.1) by elementary analysis. No proof uses a computation. The exact counts to $n = 100$ cannot confirm the constant κ (Remark 6.1). The write adds Remark 1.4 (the OEIS entry, quoted; its upper bound is exponentially loose), Remark 6.1, Remark 7.1 (the inverse against the transseries volume) and Section 10 (open questions).

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Scope and priority. All logarithms are natural. The graph model is unweighted and labelled; graphs are not counted once per compatible topological ordering. The mathematical statements below follow from the cited probability lemmas, whose hypotheses are verified explicitly. A bounded literature search found no equivalent application to this model. This does not establish first-ever priority. In particular, Steinsky’s 2003 paper on bounded indegree DAG coding was not available for a full-text inspection, so its priority implications remain unresolved. Section 9 records this limitation separately from the proof.

1 The graph model and main results

A simple directed graph has no loops and at most one edge of each orientation between two different vertices. A directed acyclic graph (DAG) has no directed cycle; in particular it cannot contain both orientations of a pair. Write $a_{n,c}$ for the number of such graphs on $[n] = \{1, \dots, n\}$ in which each vertex has at most c incoming edges. The labels are part of the object. Put $a_{0,c} = 1$, and abbreviate $a_n = a_{n,2}$. The first values are

$$a_0, a_1, a_2, a_3, a_4, a_5 = 1, 1, 3, 25, 443, 13956.$$

The sequence is A397711 [3]. Exact counting for fixed indegree bounds is established prior art; see Allen et al. [1], Theorem 18 and Table 1, and Steinsky [2].

Let Ai be the Airy function and let

$$\text{Ai}(-\zeta) = 0, \quad \zeta = 2.338107410459767\dots, \quad c_A = \frac{\zeta}{2^{1/3}}, \quad \kappa = 3c_A. \quad (1.1)$$

Here $-\zeta$ is the largest real zero, and $\kappa = 5.567271244467717\dots$. Decimal values are supplied for orientation; the exact constants are defined by (1.1).

Theorem 1.1 (The first logarithmic correction). *As $n \rightarrow \infty$,*

$$\log a_n = 2n \log n - n - \kappa n^{1/3} + o(n^{1/3}). \quad (1.2)$$

Equivalently,

$$a_n = \left(\frac{n^2}{e}\right)^n \exp\{-\kappa n^{1/3} + o(n^{1/3})\}, \quad \lim_{n \rightarrow \infty} \frac{a_n^{1/n}}{n^2} = \frac{1}{e}. \quad (1.3)$$

The error in (1.2) is only $o(n^{1/3})$. Formula (1.3) is therefore not a ratio equivalent to an explicitly given sequence. An unproved logarithmic term could still change a power prefactor, and an unproved constant term could still change an amplitude. No such refinement is inferred from this theorem or from finite data.

Theorem 1.2 (Threshold inversion). *For real $x \rightarrow \infty$, define*

$$N(x) = \min\{n \geq 1 : a_n \geq x\}, \quad y = \log x, \quad b = \frac{y}{2W_0(y/(2\sqrt{e}))}, \quad (1.4)$$

where W_0 is the positive real branch of Lambert's W function on positive arguments. Then

$$N(x) = b + \frac{\kappa b^{1/3}}{2 \log b + 1} + o\left(\frac{b^{1/3}}{\log b}\right). \quad (1.5)$$

The displacement in (1.5) is positive. The negative Airy term lowers the count at a given n , so a larger index is needed to reach x . The allowed error scale grows; the theorem supplies neither eventual exact rounding nor an additive $o(1)$ inverse error.

Theorem 1.3 (Every fixed indegree bound). *For each fixed integer $c \geq 2$, as $n \rightarrow \infty$,*

$$\begin{aligned} \log a_{n,c} &= cn \log n - [c - 1 + \log((c - 1)!)]n \\ &\quad - \kappa(c - 1)^{2/3} n^{1/3} + o(n^{1/3}). \end{aligned} \quad (1.6)$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{a_{n,c}^{1/n}}{n^c} = \frac{e^{-(c-1)}}{(c-1)!}. \quad (1.7)$$

The error term and its starting index may depend on c . No uniformity is asserted when c grows with n .

The proof is organized around a potential-transfer lemma for the Poisson-minus-one walk. The exact counting identities come first so that the factorials, interval endpoints, and area convention can all be checked before any limiting argument is applied.

Remark 1.4 ([write] The OEIS entry A397711, 6 October 2026). The live entry (revision #9, 12 July 2026, read on 6 October 2026; the revision the source recorded) by Tyler Satchel Orden (5 July 2026), offset 0, is named

Number of acyclic digraphs (or DAGs) on n labeled nodes in which every node has in-degree at most 2.

(quoted verbatim). Its model is the one of Section 1: labelled, every graph counted once, no loops and no pair joined in both directions (a DAG cannot contain both). Its 17 displayed terms ($0 \leq n \leq 16$) are those frozen in `data/oeis_prefixes.json`, and its b-file ($0 \leq n \leq 100$) equals the column $c = 2$ of `data/counts.json` term by term (checked at the write, which also checked that the entry's Python program, a composition-by-height formula the source does not use, agrees with the sink recurrence (2.2) for $n \leq 40$). The entry states no asymptotic formula. Its one inequality,

$$a(n) \leq (1 + (n-1) + \text{binomial}(n-1, 2))^n.$$

is the trivial bound $a_n \leq g(n-1)^n$ of (2.1) (each vertex chooses at most two parents among the other $n-1$). [write] *By Theorem 1.1 it is exponentially loose:*

$$\log a_n - n \log g(n-1) = -(1 - \log 2)n - \kappa n^{1/3} + o(n^{1/3}).$$

Indeed $g(n-1) = (n^2 - n + 2)/2$, so $n \log g(n-1) = 2n \log n - n \log 2 + n \log(1 - n^{-1} + 2n^{-2})$, and $n \log(1 - n^{-1} + 2n^{-2}) = -1 + O(n^{-1})$; subtract this from (1.2). The entry's comments on residues,

`a(p) == 1 (mod p)` for every prime p : the cyclic group C_p acts on the set of such DAGs by relabeling, and the only fixed DAG is the empty one.

Conjecture (verified for $n \leq 100$): `a(n) == 1 (mod n)` for all odd $n \geq 3$, and `a(n) == n/2 + 1 (mod n)` for all even $n \geq 4$.

(quoted verbatim), are outside this report, which proves nothing about them; the write only confirmed that the shipped counts satisfy both congruences of the conjecture for $3 \leq n \leq 100$. This is recorded here only; nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 157 of the session bundle that ProveIt's docs/incoming intake processed as batch 108: the archive `Bounded_Indegree_DAGs_Airy_Asymptotics_and_Inverses_Source.zip` (640,081 bytes, SHA-256 8686b71deb0d...a9db989343e, 14 files, 13 at the archive root and one in `data/`, no wrapper directory), arrival commit 60f54ea06, placed in 602e5bd0f. The text is the delivered `Report157.tex` (897 lines, dated 3 October 2026), shipped as `article.tex`. The delivered 15-page PDF and the checksum manifest `SHA256SUMS` (13 entries, verified at

placement and again at the write) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data; its delivered README used generic /tmp example paths, which the report README does not reproduce. It is a single-source report, so no merge choice was made. The write prefixed the 76 delivered labels with `bid:` and updated the 58 references to them; it added the note after the abstract, this subsection, Remarks 1.4, 6.1 and 7.1, the notes marked [write] in Sections 4, 8 and 9, and Section 10. No statement, proof or number of the source was changed: each added remark is the last statement of its section, the added section follows the last one, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A397711 [3], its b-file and the mirror file `seq/A397/A397711.seq` (still revision #9; Remark 1.4).
- Mallein [6], arXiv:1307.4496v3, the assumptions (2.12)–(2.13) and Lemmas 2.6 and 2.7 with equations (2.15) and (2.19), on pp. 11 and 13; the downloaded PDF has the SHA-256 recorded by the source in `data/SOURCE_PROVENANCE.json`. What they state, and what is used, is in the note at the end of Section 4.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:def:three-inverses`, `p0:thm:staircase` and `p0:prop:operator-form`, read for Remark 7.1.
- The neighbouring reports named below, at the passages cited.
- Not read by the write: Steinsky [2] (the source read its publisher abstract only), Allen et al. [1], Kung and Yan [4], Yan [5], Chen and Pearl [7], Yin [8] and Mallein’s journal version and author manuscript. The statements made about them in Sections 2 and 9 are the source’s; `data/SOURCE_PROVENANCE.json` records what the source inspected, with SHA-256 fingerprints.

What was checked at intake. The batch-108 dossier read the manuscript in full and found no error. It rechecked the sink recurrence, the normalization (3.3) (summation by parts, $d_i = i - 1$), the block comparisons (5.8), the identities (5.10) and (5.23), the order of limits, the Brownian eigenvalue of Section 4, the forced terminal descent (probability exactly e^{-L_n}), the algebra of Theorem 1.3 and the proof of Theorem 1.2, all modulo Mallein’s lemmas; it reran the companion on copies (note at the end of Section 8). The write rechecked the same proofs line by line and found no error; it also checked the grid bounds $T \leq A_0 < rT$ and (5.6), the interval containments (5.16)–(5.18), the initial segment (5.19) (a nonnegative path ending in I_0), the increments (6.3) and the expansion of Theorem 1.3, read Mallein’s two lemmas, recomputed the counts by its own code against the OEIS b-file, and computed the finite-data diagnostics of Remark 6.1. Finite computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository named A397711 or bounded-indegree DAGs, and none does outside this report. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. Two neighbours in `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/` share its Airy mechanism; neither proves anything about DAGs. (1) `a082161-airy-amplitudes` proves stretched exponentials $e^{3zn^{1/3}}$ with the *same* Airy zero $z = -\zeta$ for relaxed and compacted binary trees and minimal acyclic automata, with positive amplitudes and expansions to every fixed order, by Jacobi-matrix methods; its coefficient of $n^{1/3}$ is $3z = -3\zeta$, against $-\kappa = -3\zeta/2^{1/3}$ here ($c = 2$) and $-\kappa(c - 1)^{2/3}$ in general, and the constant its Parts I, II and IV call $a = 2^{-1/3}z$ is $-c_A$ of (1.1). Its data file `data/48-dfa-A331120.seq` (an OEIS snapshot of A331120) cites Priez’s enumeration of minimal acyclic automata “via generalized parking functions”; this report uses the generalized (u -)parking functions of Kung and Yan (Section 2), while `a082161`’s own proofs use no parking functions. Nothing here reaches its precision: no amplitude, no power prefactor, no second term (Question 5). (2) `a238873-diagonal-partitions`, Part IV (bundle Report 160), uses the *same* Mallein Lemmas 2.6 and 2.7, equations (2.15) and (2.19), for the $n^{1/6}$ Airy term of superdiagonal partitions (its Section `dgp:airy:sec:source`, whose write note records what the two lemmas state). There the walk is $1 - \text{Geom}(1/2)$, with $\sigma^2 = 2$ and the multiplier $(\sigma^2/2)^{1/3} = 1$; here it is $\text{Poisson}(1) - 1$, with $\sigma^2 = 1$ and

the multiplier $2^{-1/3}$, which is the $2^{-1/3}$ in c_A . Its tilt is called q ; here the tilt is h and q_i is the potential. Both reports apply the lemmas only at fixed tilts and fixed scaled intervals and sandwich a non-constant potential between them; this report does so blockwise on a geometric grid (Section 5), a238873 with one tilt sequence $q_m \rightarrow b$. Further afield, a182220-source-boundary counts *extensional* acyclic digraphs, a different class, and a089479-fixed-permanent-matrices cites the unbounded labelled DAG count A003024 in its bibliography. The Lambert inverse of Section 7 is compared with the transseries volume in Remark 7.1.

[write] What is not claimed, collected from the source. The report proves logarithmic asymptotics only: no ratio equivalent, power prefactor, amplitude, $O(\log n)$ remainder or all-orders expansion; the $O(\log n)$ of the separate product calculation does not improve the final remainder. No uniformity is asserted when c grows with n ; constants and onsets may depend on c . The inverse has a growing permitted error, no eventual exact rounding, no additive $o(1)$ error, no explicit onset and no certified finite-input bracket. The model is the unweighted labelled one, not the order-weighted model of Chen and Pearl. Exact enumeration, the general-vector parking identity and the homogeneous Airy estimates are prior inputs, and the mechanisms are not claimed as new; no new graph-to-parking bijection is claimed. The case $c = 1$ is the forest formula, with no Airy claim. Steinsky’s 2003 paper was not read in full (“This unresolved full-text priority check remains important”), with no assertion that it lacks a relevant asymptotic; the literature search was bounded and is “not a proof of global priority”. Finite checks support formulas and code and prove neither an asymptotic rate nor priority; PDF byte identity is toolchain-specific. No OEIS entry was edited. The write adds: the finite data to $n = 100$ neither confirm nor refute the constant κ (Remark 6.1), and the instance statements of Remark 7.1 are about the Lambert core only.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed; κ_{vol} , d_{vol} , L_{vol} are the write’s names for symbols of the transseries volume).

Symbol	Meaning here (where)	Not to be read as
$c, c_A, C_r, C_{T,r,q}$	indegree bound; $c_A = \zeta/2^{1/3}$ (1.1); constants in Section 5	one another; a082161’s amplitude γ
ζ, κ	$-\zeta$ the largest zero of Ai; $\kappa = 3c_A = 5.5673\dots$	a082161’s $z = -\zeta$ and its $a = 2^{-1/3}z = -c_A$; κ_{vol} of p0:prop:factorial-core
$N(x), N$	threshold index (1.4); grid horizon $N = n$ or $n - L_n$ (Section 5)	each other
$b, b_i, [a, b]$	inverse base (1.4); sorted preferences (Section 2); scaled incoming interval (5.16)	each other; a_n
$q_i, h, h_{\pm}$	potential (3.2); homogeneous tilt (4.1), (5.7)	a238873’s tilt q (here h)
d_i, u_i, L_n	increments (2.6); parking bounds (2.4); tail length (5.12)	$d_{\text{vol}}, L_{\text{vol}}$ of p0:prop:factorial-core
$r, \rho, \gamma, D(r)$	grid ratio; comparison slack; $(r/(r-1))^{1/3}$; grid constant (5.6)	Euler’s constant γ
$K_{m,h}, J, I_j$	kernel (4.1); endpoint set, number of blocks; interfaces	each other (J is both, as in the source)

2 Exact enumeration and generalized parking functions

2.1 Inclusion and exclusion over sinks

Define, with $\binom{m}{j} = 0$ when $j > m$,

$$g_c(m) = \sum_{j=0}^c \binom{m}{j}, \quad g(m) = g_2(m) = 1 + m + \binom{m}{2} = 1 + \frac{m(m+1)}{2}. \quad (2.1)$$

Proposition 2.1 (Sink recurrence). *For every integer $n \geq 1$,*

$$a_{n,c} = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} g_c(n-k)^k a_{n-k,c}. \quad (2.2)$$

Proof. Every nonempty finite DAG has a sink. Fix a set K of k labelled vertices and require all of them to be sinks. There are no edges inside K or from K to its complement. The complement supports any allowable DAG, giving $a_{n-k,c}$ choices. Each sink chooses its parent set independently from the $n-k$ complementary vertices, with at most c parents, giving $g_c(n-k)^k$ choices. These choices create no directed cycle. Inclusion and exclusion over the n sink events proves (2.2). $\square$

This direct derivation fixes the normalization and model. It does not assert that bounded-indegree DAG enumeration itself is new. In particular, one must not multiply the count by a number of topological orders; that would give a different weighted model.

2.2 General vector parking bounds

For a nondecreasing positive integer vector $u_1, \dots, u_n$, a u -parking preference word is a labelled word of n positive integers whose sorted values $b_1 \leq \dots \leq b_n$ satisfy $b_i \leq u_i$. Let $P_n(u_1, \dots, u_n)$ count these words and put $P_0 = 1$. Kung and Yan [4], Corollary 5.3 and Theorem 5.4, give the identity

$$x^n = \sum_{j=0}^n \binom{n}{j} P_j(u_1, \dots, u_j) (x - u_{j+1})^{n-j}. \quad (2.3)$$

The last term is P_n , so its apparent use of u_{n+1} is harmless. The counting decomposition initially proves this for every sufficiently large integer x . Both sides are polynomials in x , hence it is a polynomial identity and may be evaluated at $x = 0$.

Take

$$u_i = g_c(i-1), \quad u_0 = 0. \quad (2.4)$$

Moving the $j = n$ term of (2.3) to the other side and putting $k = n - j$ gives exactly (2.2). The initial values agree, so induction yields

$$a_{n,c} = P_n(g_c(0), \dots, g_c(n-1)). \quad (2.5)$$

This is an equality of counts obtained from established parking theory; no new explicit graph-to-parking bijection is required or claimed.

2.3 Occupancies and the barrier

Let

$$d_i = u_i - u_{i-1} \quad (i \geq 1). \quad (2.6)$$

In the present applications every d_i is positive. For a preference word, let x_i be the number of its letters in the integer interval $(u_{i-1}, u_i]$. The sorted parking condition is equivalent to

$$x_i \geq 0, \quad \sum_{i=1}^n x_i = n, \quad \sum_{j=1}^i x_j \geq i \quad (1 \leq i \leq n). \quad (2.7)$$

Indeed, $b_i \leq u_i$ holds exactly when at least i letters lie at or below u_i . Once the occupancies are fixed, there are $n!/\prod_i x_i!$ ways to allocate the labelled positions and $\prod_i d_i^{x_i}$ ways to choose their preference values. Therefore

$$a_{n,c} = n! \sum_{\mathbf{x} \text{ satisfying (2.7)}} \prod_{i=1}^n \frac{d_i^{x_i}}{x_i!}. \quad (2.8)$$

For $c = 2$,

$$u_i = 1 + \binom{i}{2}, \quad d_1 = 1, \quad d_i = i - 1 \quad (i \geq 2), \quad \prod_{i=1}^n d_i = (n-1)!. \quad (2.9)$$

The first two increments are both one. This initial duplication will make $q_1 = 0$ below; shifting it would change an exact identity.

3 The exact Poisson excursion representation

Let $X_1, X_2, \dots$ be independent Poisson random variables of mean one, and define

$$S_0 = 0, \quad S_i = \sum_{j=1}^i (X_j - 1), \quad E_n = \{S_i \geq 0 \ (1 \leq i \leq n), \ S_n = 0\}. \quad (3.1)$$

Then (2.7) is exactly the event E_n , under $x_i = X_i$. For positive d_i , put

$$q_0 = 0, \quad q_i = \log(d_{i+1}/d_i) \quad (i \geq 1), \quad Z_n(q) = \mathbb{E} \left[\exp \left\{ - \sum_{i=0}^{n-1} q_i S_i \right\}; E_n \right]. \quad (3.2)$$

The semicolon denotes multiplication by the event indicator.

Proposition 3.1 (Exact normalization). *For every integer $n \geq 1$, with the increments (2.6) and the potential (3.2),*

$$a_{n,c} = n! e^n \left(\prod_{i=1}^n d_i \right) Z_n(q). \quad (3.3)$$

In particular, for $c = 2$,

$$q_0 = q_1 = 0, \quad q_i = \log \frac{i}{i-1} \quad (i \geq 2), \quad (3.4)$$

$$a_n = n!(n-1)! e^n Z_n(q). \quad (3.5)$$

Proof. For any fixed occupancy vector, its Poisson probability is $e^{-n}/\prod_i x_i!$. Equation (2.8) consequently equals $n! e^n \mathbb{E}[\prod_i d_i^{X_i}; E_n]$. On E_n , summation by parts gives

$$\begin{aligned} \sum_{i=1}^n (X_i - 1) \log d_i &= \sum_{i=1}^n (S_i - S_{i-1}) \log d_i \\ &= S_n \log d_n - S_0 \log d_1 - \sum_{i=1}^{n-1} S_i \log \frac{d_{i+1}}{d_i} \\ &= - \sum_{i=1}^{n-1} q_i S_i. \end{aligned} \quad (3.6)$$

Factoring out $\prod_i d_i$ proves (3.3). Equation (2.9) gives the specialization. $\square$

The area is charged at the left endpoint of each step, from $i = 0$ to $i = n - 1$. Its first term vanishes because $S_0 = 0$. Its last term is $q_{n-1}S_{n-1}$, not q_nS_n . This convention matches the probability input below and prevents interface costs from being double-counted. For (3.4), $q_i \geq 0$ and $iq_i \rightarrow 1$.

4 The homogeneous Airy estimates used

The external probability input is Mallein [6], Lemma 2.6, equation (2.15), and Lemma 2.7, equation (2.19), in the arXiv version 3 and corresponding author manuscript. The assumptions are given there in equations (2.12)–(2.13). We state precisely the homogeneous variance-one specialization that is used here.

When the walk starts at $x \in \mathbb{Z}_{\geq 0}$, write $\mathbb{E}_x$ for expectation and still write S_j for its absolute height. For $m \geq 1$, $h > 0$, and $J \subseteq [0, \infty)$ define the killed area-tilt kernel

$$K_{m,h}(x, J) = \mathbb{E}_x \left[\exp \left\{ -\frac{h}{m} \sum_{j=0}^{m-1} S_j \right\}; S_j \geq 0 \ (0 \leq j \leq m), \ S_m \in J \right]. \quad (4.1)$$

Every endpoint interval below is intersected with $\mathbb{Z}$ when used for this walk.

Proposition 4.1 (Homogeneous bounds from Mallein). *For each fixed $h > 0$,*

$$\limsup_{m \rightarrow \infty} m^{-1/3} \log \sup_{x \in \mathbb{Z}_{\geq 0}} K_{m,h}(x, [0, \infty)) \leq -c_A h^{2/3}. \quad (4.2)$$

For each fixed $h > 0$ and fixed $0 < a < b$, $0 < a' < b'$,

$$\liminf_{m \rightarrow \infty} m^{-1/3} \log \inf_{x \in [am^{1/3}, bm^{1/3}] \cap \mathbb{Z}} K_{m,h}(x, [a'm^{1/3}, b'm^{1/3}]) \geq -c_A h^{2/3}. \quad (4.3)$$

The source treats independent centered increments with continuous positive variance profile and a uniform exponential moment of their absolute values. For $Y = X_1 - 1$,

$$\mathbb{E}Y = 0, \quad \mathbb{E}Y^2 = 1, \quad \mathbb{E}e^{\mu|Y|} < \infty \quad \text{for every fixed } \mu > 0.$$

Thus the homogeneous array has variance profile identically one and satisfies every required moment condition. Neither symmetry nor a nonlattice law is assumed in the cited estimates. The lower estimate uses an interval of terminal heights, not an exact endpoint. The exact zero endpoint in our application will be supplied separately.

The normalization of c_A can also be checked at the Brownian level. The killed operator $\frac{1}{2}d^2/dx^2 - hx$ on the positive half-line has eigenfunction $\text{Ai}((2h)^{1/3}x - \zeta)$ with eigenvalue $-\zeta h^{2/3}/2^{1/3}$. This verifies the variance-one convention; it is not a replacement proof of the random-walk rare-event bounds.

Remark 4.2 (The quantifier needed for many blocks). Fix h , the endpoint parameters when needed, and $\epsilon > 0$. Equations (4.2)–(4.3) give a single integer m_0 such that the corresponding bounds

$$K \leq e^{(-c_A h^{2/3} + \epsilon)m^{1/3}}, \quad K \geq e^{(-c_A h^{2/3} - \epsilon)m^{1/3}}$$

hold for every integer $m \geq m_0$, with the indicated supremum or infimum. This elementary consequence of limsup and liminf is the uniformity used below. We do not vary h or the scaled endpoint intervals with the total horizon, and do not require a uniform theorem over such varying parameters.

[write] Mallein's two lemmas as the write read them, 6 October 2026. The write read the two cited statements in arXiv:1307.4496v3 [6] (journal version and author manuscript not compared), as the write of `a238873-diagonal-partitions` did for its Part IV. For a triangular array of independent centred variables $X_{n,k}$ with $\mathbb{E}[X_{n,k}^2] = \sigma_{k/n}^2$, σ continuous and positive (2.12), and a uniform exponential moment $\sup_{n,k} \mathbb{E}e^{\lambda|X_{n,k}|} < \infty$ for some $\lambda > 0$ (2.13), Lemma 2.6, equation (2.15) (p. 11), bounds $\limsup_n n^{-1/3} \log \sup_{x \in \mathbb{R}} \mathbb{E}_x[e^{-\frac{h}{n} \sum_{j=0}^{n-1} S_j}; S_j \geq 0, j \leq n]$ above by $\frac{\alpha_1}{2^{1/3}}(h\sigma)^{2/3}$, and Lemma 2.7, equation (2.19) (p. 13), bounds the $\liminf$ of the infimum over starting points $x \in [an^{1/3}, bn^{1/3}]$ of the same expectation with the extra indicator $S_n \in [a'n^{1/3}, b'n^{1/3}]$ below by $\frac{\alpha_1}{2^{1/3}}(h\bar{\sigma})^{2/3}$, for $h > 0$, $0 < a < b$, $0 < a' < b'$; here α_1 is the largest zero of Ai (so $\alpha_1 = -\zeta$), and $\sigma, \bar{\sigma}$ are the minimum and maximum of σ . *What is used.* Only these two inequalities, (2.15) and (2.19), for the homogeneous array $X_{n,k} = X_k - 1$ with X_k Poisson of mean one: then $\sigma \equiv 1$, (2.13) holds for every λ , and both constants equal $-\zeta h^{2/3}/2^{1/3} = -c_A h^{2/3}$. Mallein's bounds with the supremum over all real x and the infimum over the whole real interval imply (4.2) (supremum over integers $x \geq 0$) and (4.3) (infimum over the integers of the interval); the event $S_j \geq 0$ for $j \leq n$ and the area $\sum_{j=0}^{n-1} S_j$ are those of (4.1). Neither (2.16) nor (2.20) (walks confined to a strip) nor any time-inhomogeneous case is used: the inhomogeneity $q_i \approx \beta/i$ is handled in Section 5 by fixed homogeneous tilts on each block. The write did not verify Mallein's proofs, and neither did the intake; the report's analytic theorems hold *modulo* these two published lemmas. In `a238873-diagonal-partitions` the same lemmas are applied with $\sigma^2 = 2$ (note in its Section `dgp:airy:sec:source`).

5 Transferring the estimates to a singular potential

Lemma 5.1 (A reciprocal potential). *For the walk (3.1), let $(q_i)_{i \geq 0}$ be finite real numbers such that $q_i \geq 0$ and*

$$iq_i \longrightarrow \beta > 0. \quad (5.1)$$

Then

$$\log Z_n(q) = -3c_A \beta^{2/3} n^{1/3} + o(n^{1/3}). \quad (5.2)$$

The value of q_0 is immaterial because $S_0 = 0$.

The potential is singular after rescaling time near zero. A direct substitution into a smooth bounded-potential theorem would therefore need justification. The proof below uses only the homogeneous bounds, with a fixed initial cutoff and a geometric partition.

5.1 Geometric grid and rounding control

Fix $r > 1$ and a large constant $T > 1$. For integers $N \rightarrow \infty$, put

$$J = \lfloor \log_r(N/T) \rfloor, \quad A_j = Nr^{j-J}, \quad t_j = \lfloor A_j \rfloor \quad (0 \leq j \leq J), \quad m_j = t_j - t_{j-1}. \quad (5.3)$$

In particular, $T \leq A_0 < rT$, $T - 1 \leq t_0 < rT$, and $t_J = N$. Writing $M_j = A_j - A_{j-1}$ gives

$$|m_j - M_j| < 1, \quad M_j = (r - 1)A_0 r^{j-1}. \quad (5.4)$$

For fixed r and sufficiently large T , all m_j are positive and bounded below by a constant times Tr^{j-1} . Uniformly over all the blocks,

$$\frac{m_j}{t_{j-1}} = (r - 1) + O(T^{-1}), \quad \frac{m_j}{t_j} = (1 - r^{-1}) + O(T^{-1}). \quad (5.5)$$

The constants here may depend on r .

The cube-root sum has the limit

$$N^{-1/3} \sum_{j=1}^J m_j^{1/3} \longrightarrow D(r) = \frac{(1 - r^{-1})^{1/3}}{1 - r^{-1/3}}. \quad (5.6)$$

To verify rounding does not affect this limit, the mean value theorem and (5.4) give

$$|m_j^{1/3} - M_j^{1/3}| \leq C_r M_j^{-2/3}$$

when T is sufficiently large. The right side is summable over the geometric grid, with a sum bounded independently of N . The exact geometric sum is

$$N^{-1/3} \sum_{j=1}^J M_j^{1/3} = (1 - r^{-1})^{1/3} \sum_{k=0}^{J-1} r^{-k/3},$$

which tends to $D(r)$. This proves (5.6).

The minimum block length is not asserted to tend to infinity with N . Instead, T is chosen sufficiently large once the fixed parameters and required length thresholds have been selected. It then remains fixed while $N \rightarrow \infty$.

5.2 Comparing the potential on each block

Fix also $\rho \in (0, 1)$ and set

$$h_- = (1 - \rho)\beta(1 - r^{-1}), \quad h_+ = (1 + \rho)\beta(r - 1). \quad (5.7)$$

After increasing T , condition (5.1) and (5.5) imply simultaneously on every block

$$\frac{h_-}{m_j} \leq q_i \leq \frac{h_+}{m_j} \quad (t_{j-1} \leq i < t_j). \quad (5.8)$$

For example, first bound iq_i between $(1 - \delta)\beta$ and $(1 + \delta)\beta$ for all indices above a fixed cutoff. On the block use $i \leq t_j$ for the lower bound and $i \geq t_{j-1}$ for the upper bound. Choose $\delta > 0$ small in terms of r, ρ , then enlarge T to absorb the ratio errors. No monotonicity of q_i is needed. Nonnegativity permits the kernel comparisons on nonnegative paths and removal of the initial cost in the upper bound.

5.3 Upper bound

Take $N = n$. Remove the final requirement $S_n = 0$ and the nonnegative area cost before t_0 . The remaining initial distribution on nonnegative heights has total mass at most one. The Markov property and (5.8) yield

$$Z_n(q) \leq \prod_{j=1}^J \sup_{x \in \mathbb{Z}_{\geq 0}} K_{m_j, h_-}(x, [0, \infty)). \quad (5.9)$$

For fixed $r, \rho, \epsilon > 0$, choose T still larger so that every m_j exceeds the single threshold in Remark 4.2. Consequently,

$$\log Z_n(q) \leq (-c_A h_-^{2/3} + \epsilon) \sum_{j=1}^J m_j^{1/3}.$$

Take $n \rightarrow \infty$, then $\epsilon \downarrow 0$, then $\rho \downarrow 0$, while r remains fixed. Formula (5.6) and the algebraic identity

$$(1 - r^{-1})^{2/3} D(r) = 1 + r^{-1/3} + r^{-2/3} \quad (5.10)$$

give

$$\limsup_{n \rightarrow \infty} n^{-1/3} \log Z_n(q) \leq -c_A \beta^{2/3} (1 + r^{-1/3} + r^{-2/3}). \quad (5.11)$$

5.4 Lower bound interfaces with explicit slack

Fix in addition $\eta > 0$, and reserve a short terminal interval by setting

$$L_n = \lceil 2\eta n^{1/3} \rceil, \quad N = n - L_n. \quad (5.12)$$

Use (5.3) with this N . At ordinary interfaces require

$$S_{t_j} \in I_j = [t_j^{1/3}, 2t_j^{1/3}] \quad (0 \leq j < J), \quad (5.13)$$

and at the last interface require

$$S_N \in I_J^{(\eta)} = [\eta N^{1/3}, 2\eta N^{1/3}]. \quad (5.14)$$

Only sufficiently large n , hence $J \geq 1$, is relevant.

The following fixed intervals make the choice of parameters transparent. Put

$$\alpha = (r - 1)^{-1/3}, \quad \gamma = \left(\frac{r}{r - 1} \right)^{1/3}. \quad (5.15)$$

By (5.5), after increasing T the scaled incoming sets $I_{j-1}/m_j^{1/3}$ all lie in

$$[a, b] = [\alpha/2, 3\alpha]. \quad (5.16)$$

All ordinary scaled outgoing sets $I_j/m_j^{1/3}$ contain the fixed interval

$$[a', b'] = [5\gamma/4, 7\gamma/4], \quad (5.17)$$

and the last set $I_J^{(\eta)}/m_J^{1/3}$ contains

$$[a'_\eta, b'_\eta] = [5\eta\gamma/4, 7\eta\gamma/4]. \quad (5.18)$$

Indeed, the limiting ordinary outgoing interval is $[\gamma, 2\gamma]$, and the last one is $[\eta\gamma, 2\eta\gamma]$, so these choices have strict positive slack. These intervals depend on the fixed r, η only. We can therefore select the two lower-bound length thresholds first and subsequently increase T to exceed their maximum. Increasing T preserves the required containments, so there is no circular choice.

5.5 A finite initial segment

For any possible t_0 , choose an integer $H \in [t_0^{1/3}, 2t_0^{1/3}]$. Force $X_1 = H + 1$ and $X_2 = \dots = X_{t_0} = 1$. This path is nonnegative, ends at height H , and has weighted probability

$$\frac{e^{-t_0}}{(H + 1)!} \exp \left\{ -H \sum_{i=1}^{t_0-1} q_i \right\}. \quad (5.19)$$

For fixed T, r , both t_0 and the selected H take values in finite sets. Because each q_i is finite, the minimum of (5.19) over these possibilities is a positive constant $C_{T,r,q}$. It may be very small as T increases, but T remains fixed in the limit in n , so $n^{-1/3} \log C_{T,r,q} \rightarrow 0$.

5.6 Multiplying the lower kernels

Apply (4.3) at $h = h_+$, with incoming interval (5.16), and with the ordinary or last outgoing interval as appropriate. For fixed r, ρ, η, ϵ , enlarge T so all lengths exceed both thresholds. The actual block transition mass is at least

$$\exp\{(-c_A h_+^{2/3} - \epsilon)m_j^{1/3}\}. \quad (5.20)$$

The reason is that (5.8) bounds the actual potential above by h_+/m_j , so its weight is at least the homogeneous weight.

For a given incoming height in I_{j-1} , the selected outgoing interval is contained in I_j (or in $I_j^{(\eta)}$ for the last block). Every resulting point of I_j is covered by the incoming infimum for the next block. The Markov property thus multiplies (5.20) without an endpoint matching loss. Each transition mass already sums over its terminal lattice points; there is no division by an interval length or additional per-state factor. The left-endpoint area convention assigns each interface cost exactly once, to its outgoing block.

Although the number of blocks grows with n , the total logarithmic error in their product is only $\epsilon \sum_j m_j^{1/3}$. After division by $N^{1/3}$ it tends to $\epsilon D(r)$. This uses the all- $m \geq m_0$ bound, not an unjustified exchange of a blockwise limit with an increasing product.

5.7 Closing the excursion at zero

At time N let the integer height be $s \in I_j^{(\eta)}$. Then $s \leq 2\eta N^{1/3} \leq L_n$. For the remaining L_n steps force $X_i = 0$ for the first s steps and $X_i = 1$ for the remaining steps. The walk descends by one to zero, then stays at zero. Both prescribed outcomes have probability e^{-1} , so this conditional event has probability exactly e^{-L_n} , independently of s .

Throughout the tail the height is at most L_n . Since $N/n \rightarrow 1$, condition (5.1) gives $\max_{N \leq i < n} q_i = O(n^{-1})$. Therefore its extra area satisfies

$$\sum_{i=N}^{n-1} q_i S_i \leq L_n^2 \max_{N \leq i < n} q_i = O(n^{-1/3}). \quad (5.21)$$

The weighted tail probability is at least $\exp\{-L_n - O(n^{-1/3})\}$. This step uses the special support of the Poisson-minus-one walk; Lemma 5.1 is asserted precisely for that walk, which is also the walk used for all fixed c .

Combining the initial segment, all blocks, and the terminal tail gives

$$\log Z_n(q) \geq \log C_{T,r,q} + (-c_A h_+^{2/3} - \epsilon) \sum_{j=1}^J m_j^{1/3} - L_n - O(n^{-1/3}). \quad (5.22)$$

First take $n \rightarrow \infty$. Then send $\epsilon, \rho \downarrow 0$, with r, η fixed. Using $N/n \rightarrow 1$, (5.6), and

$$(r-1)^{2/3} D(r) = 1 + r^{1/3} + r^{2/3}, \quad (5.23)$$

we obtain

$$\liminf_{n \rightarrow \infty} n^{-1/3} \log Z_n(q) \geq -c_A \beta^{2/3} (1 + r^{1/3} + r^{2/3}) - 2\eta. \quad (5.24)$$

Finally send $\eta \downarrow 0$ and $r \downarrow 1$ in (5.11) and (5.24). Both bounds tend to $-3c_A \beta^{2/3}$, proving Lemma 5.1.

Remark 5.2 (Order of limits). The fixed parameters r, ρ, η, ϵ are selected first, then T is fixed sufficiently large, and only then n tends to infinity. At each fixed r , the error parameter ϵ tends to zero before $r \downarrow 1$. This matters because $D(r)$ diverges as $r \downarrow 1$. No quantitative rate in the homogeneous estimates is needed or claimed.

6 Completion of the DAG asymptotics

For $c = 2$, the exact potential (3.4) satisfies $iq_i \rightarrow 1$. Lemma 5.1 therefore gives

$$\log Z_n(q) = -\kappa n^{1/3} + o(n^{1/3}). \quad (6.1)$$

Using $(n-1)! = n!/n$, Stirling's formula yields

$$\begin{aligned} \log(n!(n-1)!e^n) &= 2\log(n!) - \log n + n \\ &= 2n\log n - n + \log(2\pi) + O(n^{-1}). \end{aligned} \quad (6.2)$$

Combining (3.5), (6.1), and (6.2) proves Theorem 1.1.

The two factorials cancel their logarithmic terms. This cancellation is an exact normalization check, not evidence of a missing universal prefactor. Any logarithmic contribution in $\log a_n$ beyond (6.2) must come from a sharper analysis of $Z_n(q)$; Lemma 5.1 does not provide that sharper analysis.

6.1 Extension to every fixed c

Fix an integer $c \geq 2$. Pascal's identity applied to (2.1) gives

$$d_1 = 1, \quad d_i = g_c(i-1) - g_c(i-2) = \sum_{j=0}^{c-1} \binom{i-2}{j} \quad (i \geq 2). \quad (6.3)$$

These increments are positive and nondecreasing, so $q_i = \log(d_{i+1}/d_i) \geq 0$. As $i \rightarrow \infty$ with c fixed,

$$d_i = \frac{i^{c-1}}{(c-1)!} (1 + O(i^{-1})), \quad iq_i \rightarrow c-1. \quad (6.4)$$

For the second assertion one may expand the fixed polynomial one term further: $d_i = Ai^{c-1}(1+B/i+O(i^{-2}))$ for constants $A > 0, B$. Taking a ratio at consecutive integers gives $q_i = (c-1)/i + O(i^{-2})$.

Summing logarithms in (6.4) gives

$$\log \prod_{i=1}^n d_i = (c-1)\log(n!) - n\log((c-1)!) + O(\log n). \quad (6.5)$$

Finite initial indices are absorbed in the error. Proposition 3.1, Lemma 5.1 with $\beta = c-1$, and Stirling's formula now give

$$\begin{aligned} \log a_{n,c} &= c\log(n!) + n - n\log((c-1)!) - \kappa(c-1)^{2/3}n^{1/3} + o(n^{1/3}) + O(\log n) \\ &= cn\log n - [c-1 + \log((c-1)!)]n - \kappa(c-1)^{2/3}n^{1/3} + o(n^{1/3}). \end{aligned}$$

This proves Theorem 1.3. The $O(\log n)$ error in the separate product calculation is absorbed by $o(n^{1/3})$; it does not improve the final remainder to $O(\log n)$.

For context, $c = 1$ is a different boundary case: such DAGs are rooted labelled forests with edges oriented away from the roots, and $a_{n,1} = (n+1)^{n-1}$ for $n \geq 1$. In that case $d_i = 1$ and $q_i = 0$, so $\beta = 0$. The positive- β transfer lemma is not used to claim an Airy correction there. The forest formula is a useful independent finite check of the exact recurrence.

Remark 6.1 ([write] What the exact counts to $n = 100$ can and cannot say, 6 October 2026). Put $R_n = \log a_n - (2n \log n - n) + \kappa n^{1/3}$, so that Theorem 1.1 says $R_n/n^{1/3} \rightarrow 0$. From the exact counts (the write's own sink-recurrence code, equal to `data/counts.json` and to the OEIS b-file), with $\log Z_n = \log a_n - \log(n!(n-1)!e^n)$ by (3.5):

n	10	25	50	75	100
R_n	8.332	8.799	9.165	9.386	9.546
$R_n/n^{1/3}$	3.867	3.009	2.488	2.226	2.057
$n^{-1/3} \log Z_n$	-2.561	-3.189	-3.579	-3.778	-3.907

against the limit $-\kappa = -5.5673$ of the last row. The ratio $R_n/n^{1/3}$ decreases slowly and is still about 2 at $n = 100$; R_n contains $\log(2\pi) = 1.838$ from (6.2) and an unknown, unproved lower-order part of $\log Z_n$. The constant κ cannot be confirmed from these counts, and the table is not evidence for it. Least squares fits of the exact values of $\log Z_n$ over all integers of the windows $[20, 100]$, $[40, 100]$, $[60, 100]$ (50-digit arithmetic) give the coefficient of $n^{1/3}$ as -5.502 , -5.506 , -5.509 with the basis $\{n^{1/3}, \log n, 1\}$, and -5.515 , -5.534 , -5.542 with $n^{-1/3}$ added: every value lies above $-\kappa$, by 1.2% down to 0.5%, moving towards it as the window shrinks and the basis grows. These fits presuppose an expansion of $\log Z_n$ in those functions, which is not proved (the theorem gives only $o(n^{1/3})$), so they are consistent with κ under that assumption and confirm nothing; their coefficient of $\log n$ (0.46 to 0.56) is not stable across windows and bases, and no power prefactor is inferred from it. (Wording corrected after the independent check of 6 October 2026; the first wording read “is not stable, and suggests no power prefactor”, which can be read as evidence against a power prefactor. The data do not support that reading: the fitted coefficient of $\log n$ stays near $\frac{1}{2}$, and with the coefficient of $n^{1/3}$ fixed at $-\kappa$ the same three windows give 0.542, 0.549, 0.554. Nothing is inferred from this either.) The question is Question 1.

7 Lambert W and the threshold inverse

The threshold in (1.4) exists because the counts grow without bound. They are strictly increasing for $n \geq 2$: extend any allowed graph on $[n-1]$ by vertex n as a sink, either isolated or with one of the $n-1$ old vertices as its sole parent. These n extensions are distinct and respect the indegree bound, so

$$a_n \geq n a_{n-1} \quad (n \geq 2). \quad (7.1)$$

Put $F(t) = 2t \log t - t$. If $w = W_0(y/(2\sqrt{e}))$, then $w e^w = y/(2\sqrt{e})$ and

$$b = \frac{y}{2w} = \sqrt{e} e^w, \quad \log b = \frac{1}{2} + w, \quad F(b) = 2bw = y.$$

Thus b is the exact large positive solution of the leading smooth model, and

$$F'(b) = 2 \log b + 1, \quad F''(b) = 2/b. \quad (7.2)$$

Here are explicit bracketing details for the error precision. Fix $\epsilon > 0$. Theorem 1.1 implies that for every sufficiently large integer n ,

$$F(n) - (\kappa + \epsilon)n^{1/3} \leq \log a_n \leq F(n) - (\kappa - \epsilon)n^{1/3}. \quad (7.3)$$

For any fixed real A , let

$$t_A = b + \frac{Ab^{1/3}}{F'(b)}.$$

Taylor's theorem and (7.2) give

$$F(t_A) = y + Ab^{1/3} + O\left(\frac{b^{-1/3}}{\log^2 b}\right), \quad t_A^{1/3} = b^{1/3} + O\left(\frac{b^{-1/3}}{\log b}\right). \quad (7.4)$$

Integer rounding changes $F(t_A)$ by $O(\log b)$, which is $o(b^{1/3})$. Choosing $A = \kappa - 2\epsilon$ and $A = \kappa + 2\epsilon$ in (7.3) shows, for all sufficiently large x ,

$$\lfloor t_{\kappa-2\epsilon} \rfloor < N(x) \leq \lceil t_{\kappa+2\epsilon} \rceil. \quad (7.5)$$

For the left inequality the upper count bound is below y ; for the right inequality the lower count bound is above y . The strict margins are of order $\epsilon b^{1/3}$ and dominate the Taylor and rounding errors.

Finally, $O(1) = o(b^{1/3}/\log b)$. Divide the displacement bounds by $b^{1/3}/F'(b)$ and let $\epsilon \downarrow 0$. This proves (1.5). No numerical asymptotic onset or explicit remainder constant is supplied by this proof, so it does not certify a finite-input numerical bracket for $N(x)$.

Remark 7.1 ([write] The inverse against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`; its symbols κ, d, L are written $\kappa_{\text{vol}}, d_{\text{vol}}, L_{\text{vol}}$ here, to keep them apart from κ, d_i and L_n of this report.

- (a) *The base b is an instance*, verbatim, of `p0:prop:factorial-core`, with $(\kappa_{\text{vol}}, d_{\text{vol}}, L_{\text{vol}}) := (2, -1, y)$, $y = \log x > 0$: the phase is $X(2\log X - 1) = F(X)$, the volume's v is $W_0((y/2)e^{-1/2}) = W_0(y/(2\sqrt{e})) = w$, its solution is $X = e^{v+1/2} = \sqrt{e}e^w = b$, its side condition $2\log X - 1 > 0$ is $w > 0$, which holds since $y > 0$, and its slope $\kappa_{\text{vol}}(v+1) = 2w+2 = 2\log b+1$ is $F'(b)$ of (7.2).
- (b) *$N(x)$ is the volume's staircase, but Theorem 1.2 is not an instance of `p0:thm:staircase`, only an analogue.* By $a_1 = 1 < a_2 = 3$ and (7.1), $(a_n)_{n \geq 1}$ is strictly increasing, so $N(x) = N_*(x)$ of `p0:def:three-inverses` with $n_0 = n_1 = 1$, and `p0:thm:staircase(1)` gives $N(x) = \lceil \nu(x) \rceil$ for every admissible interpolation. The proof here uses no interpolation: it brackets $N(x)$ directly at the integers, between $\lfloor t_{\kappa-2\epsilon} \rfloor$ and $\lceil t_{\kappa+2\epsilon} \rceil$ in (7.5), from two envelopes of $\log a_n$. Nor does the separation condition of `p0:thm:staircase(2)` apply: the theorem locates $N(x)$ only up to an error $o(b^{1/3}/\log b)$, not shown to be small (for fixed ϵ the bracket has width of order $\epsilon b^{1/3}/\log b \rightarrow \infty$), which is why no eventual exact rounding is claimed.
- (c) *The displacement is a first-order perturbation term, not an instance of a theorem of the volume.* With $\Phi_0 = F$, $X = b$ and $\varepsilon \varrho(t) = -\kappa t^{1/3}$, the $m = 1$ term of `p0:eq:operator-series` in `p0:prop:operator-form` is $-\varepsilon \varrho(b)/F'(b) = \kappa b^{1/3}/(2\log b + 1)$, the displacement of (1.5). That proposition is a formal series in ε for a fixed perturbation; here $\log a_n - F(n)$ is known only as $-\kappa n^{1/3} + o(n^{1/3})$ at the integers, so the agreement is formal and no higher term of the series is meaningful.

No novelty is claimed for the inversion.

8 Exact companion and reproducibility

The accompanying source package contains the editable LaTeX article, the PDF, a Python standard-library companion, tests, attributed numerical prefixes, deterministic exact output, source provenance, and controlled build and archive helpers. No journal full text or unrelated research material is bundled.

The count calculation uses the sink recurrence (2.2) with arbitrary-precision integers. Independently structured checks use the parking occupancy constraints, rational Poisson area weights, and direct small-graph enumeration. In the latter, each unordered vertex pair has three possible states: absent, one orientation, or the other. An indegree check and an acyclicity test then select precisely the required labelled simple DAGs. Small parking-word enumeration checks the weak upper-bound convention directly. The $c = 1$ forest formula supplies another independent reference case.

The exact threshold operation searches a bounded table of actual counts. If its chosen range is too short, it reports that fact rather than replacing an exact answer by the asymptotic inverse. Workload limits are software limits only and do not restrict the mathematical theorems. Finite checks support the formulas' indexing and the code's correctness; they prove neither an asymptotic rate nor scholarly priority.

The README records the exact command lines, bounds, shipped test coverage, and output policies. Validation is explicit rather than implemented with Python assertions that disappear under optimization; the suite is run both normally and with `-O`. Output is exclusive and refuses existing targets and symbolic-link traversal. These are local safety measures under a trusted-parent and nonadversarial-concurrent-environment assumption, not an operating-system sandbox or a claim of general race resistance.

For the PDF, the helper creates an exclusive clean build directory, uses a fresh LaTeX format and cache, disables shell escape, fixes PDF metadata, and repeats compilation until references stabilize. The settled log must contain no warnings or layout diagnostics. Identical bytes are expected on repeated builds within the recorded toolchain, not across arbitrary TeX versions, fonts, or platforms. In particular, the directory creation and subsequent opening require a trusted parent and no adversarial concurrent replacement.

The release manifest covers a fixed sorted payload allowlist. The ZIP uses fixed timestamps, permissions, order, and stored compression; it includes the checksum manifest, which necessarily excludes its own self-hash. A separately reported archive digest identifies the complete release. The supplied commands permit checksum validation, fresh extraction, exact-output replay, normal and optimized tests, clean PDF rebuild, and deterministic archive replay.

[write] **The package here, and the intake reruns, 6 October 2026.** “The accompanying source package”, “the README” and “the release manifest” above are the delivered ones. In this report the companion and its tests are `code/companion.py` and `code/test_companion.py`; the frozen outputs are `data/counts.json` and `data/finite_checks.json` (delivered at the archive root) and `data/oeis_prefixes.json`; the source provenance is `data/SOURCE_PROVENANCE.json`; the build and release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`. The delivered PDF and SHA256SUMS are not shipped. The shipped layout does not run as delivered: the companion reads `data/oeis_prefixes.json` beside itself (that is, `code/data/`), the tests read `counts.json` and `finite_checks.json` beside themselves, `build_pdf.py` compiles `Report157.tex` and `make_zip.py` checks a closed allowlist against SHA256SUMS; so reruns go through a fresh extraction of the archive (the report README gives the route). Its safe-output helpers need POSIX (`O_NOFOLLOW`, `os.mkfifo`). *Reruns on copies* (Windows, Python 3.14.4). At placement the dossier found: `counts -max-n 100 -max-c 5` reproduces `counts.json`; the `verify` command refuses on Windows (no `O_NOFOLLOW` in its input reader), and with only that reader replaced by a plain read (an

intake helper, not shipped) `verify` with the shipped bounds reproduces `finite_checks.json` byte for byte; `threshold -value 1000` returns `first_n 5`; of the 34 companion tests, 3 fail and 4 err, all on the POSIX output guard, and 14 are skipped as POSIX-only; 10 of the 11 release tests err (`os.mkfifo`). At the write, on a fresh extraction: `SHA256SUMS` checks 13 of 13, `counts` reproduces `counts.json` once Windows line endings are converted, and `threshold -value 1000 -c 2 -max-n 100` returns $n = 5$. No failure is mathematical. The PDF and ZIP rebuilds need the delivering TeX Live and POSIX and were not run.

9 Attribution and limits of source clearance

The argument has three distinct prior inputs. First, bounded-indegree labelled DAG enumeration is already treated by Steinsky [2] and Allen et al. [1]. Second, the general-vector parking identity is due to Kung and Yan [4]; Yan’s survey [5] provides additional context. Third, the homogeneous killed-walk area-tilt estimates are Mallein’s [6]. The proof here applies these inputs to a reciprocal potential, verifies the endpoint and block quantifiers, and derives the stated DAG logarithmic asymptotic and threshold displacement. It does not claim any of the underlying mechanisms as new.

The source preparation for this report inspected the relevant exact counting section of Allen et al., all pages of the Kung–Yan author manuscript through rendering and OCR, and the specific Mallein assumptions, lemmas, and nearby proofs used above. The crucial formula pages were checked visually because the parking manuscript’s original text extraction was damaged. The homogeneous estimates also agree between Mallein’s arXiv version 3 and author manuscript. These statements record the scope of the source check, not a claim that all surrounding literature was exhaustively reviewed.

Steinsky’s publisher abstract verifies bounded-indegree enumeration, generation, ranking, and unranking, but the full text was not read in this source pass. No assertion is made that it contains no relevant asymptotic. This unresolved full-text priority check remains important before any claim of novelty or publication priority.

Related models require care. Chen and Pearl’s bounded-indegree Bayesian-network sampling model [7] weights DAGs through compatible orders, whereas the present count gives every labelled graph weight one. General parking-function asymptotics with affine bounds, such as Yin [8], concern a different boundary from the quadratic $u_i = 1 + \binom{i}{2}$ here. Results for related graph and parking models do not by themselves identify the enumeration of arbitrary labelled simple DAGs. A bounded search found no equivalent result for this exact model, but absence from a search is not a proof of global priority.

The remaining analytic target is a quantitative expansion of the inhomogeneous excursion partition function that controls terms below $n^{1/3}$. Such a result would be needed to isolate a logarithmic prefactor, an amplitude, or further powers. None is supplied here. Similarly, no uniform extension to growing c is obtained: all constants, cutoffs, and probability parameters in the fixed- c argument are chosen after c is fixed.

[write] The OEIS record and the literature at the commit of this write, 6 October 2026. The entry was still at revision #9 (12 July 2026) on 6 October 2026 (Remark 1.4); it states no asymptotic formula. The write did not obtain Steinsky’s paper either, so the priority limitation above stands unchanged (Question 4). Mallein’s two lemmas were read in the arXiv version the source names (note in Section 4). The “remaining analytic target” and the growing- c limitation of the two preceding paragraphs are restated, with what is missing, as Questions 1 and 2.

10 Further questions and research

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The source’s Section 9 names the open targets, and its non-claims are collected in Section 1.1. The intake found no false claim in the source, and the write found none either. Question 5 is a suggestion of the intake, not a claim of the source.

1. **The remainder below $n^{1/3}$** (source: Section 9, “The remaining analytic target is a quantitative expansion of the inhomogeneous excursion partition function that controls terms below $n^{1/3}$ ”; Section 6). Proved: $\log Z_n = -\kappa n^{1/3} + o(n^{1/3})$, hence (1.2), modulo Mallein’s lemmas. Open: the next terms of $\log Z_n$ (a logarithmic term, hence a power prefactor of a_n ; a constant, hence an amplitude; further powers), and whether the remainder in (1.2) is $O(\log n)$. Missing: a quantitative version of the homogeneous estimates with rates, or a different method; Lemma 5.1 is a limit statement, and its order of limits (ϵ before $r \downarrow 1$) gives no rate. *Finite data* (intake and write, Remark 6.1): the exact counts to $n = 100$ cannot confirm the constant κ ; $R_n/n^{1/3}$ is still about 2 at $n = 100$, and fits that assume an unproved form of the remainder give coefficients within 0.5% to 1.2% of $-\kappa$, which is no confirmation. A proof of the next term, or exact counts far beyond $n = 100$ with a proved form of the remainder, would be needed.
2. **Growing indegree bounds** (source: Theorem 1.3 and Section 9, “no uniform extension to growing c is obtained”). Open: the range of $c = c(n)$ in which (1.6) holds. Missing: control of the cutoff T , of the onset of $iq_i \rightarrow c - 1$ (whose error term in (6.4) depends on c) and of the Mallein thresholds of Remark 4.2 uniformly in c .
3. **Effective onsets and a certified inverse** (source: Section 7, “No numerical asymptotic onset or explicit remainder constant is supplied by this proof, so it does not certify a finite-input numerical bracket for $N(x)$ ”; the remark after Theorem 1.2). Open: explicit $n_0(\epsilon)$ in (7.3); a certified bracket for $N(x)$ at given x ; eventual exact rounding, or an additive $o(1)$ error. Missing: effective versions of Mallein’s lemmas and of the transfer, and a remainder below $1/2$ in the inverse (Remark 7.1(b)). The companion’s threshold command searches actual counts instead.
4. **The literature boundary** (source: the scope paragraph and Section 9, “This unresolved full-text priority check remains important before any claim of novelty or publication priority”). Open: whether Steinsky [2], or other work on bounded-indegree DAGs, Bayesian-network structure counts or quadratic-bound parking functions, already contains the asymptotic. Missing: reading Steinsky’s full text (neither the source nor the write did) and a wider search.
5. **Exact Airy amplitudes** (intake suggestion, batch-108 dossier; not a claim of the source). Do the methods that give amplitudes and expansions to every fixed order for stretched exponentials $e^{3zn^{1/3}}$ in `a082161-airy-amplitudes` (whose growth orders are due to Elvey Price, Fang and Wallner) give a power prefactor and constant for A397711, or for every fixed c ? Missing: any such analysis of $Z_n(q)$ with the time-inhomogeneous potential $q_i = \log(i/(i - 1))$; whether `a082161`’s structures carry over was not examined.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit `e5152f378`) re-derived the statements marked [write] from their sources. *Remark 1.4.*

A literal brute force over all simple digraphs (each ordered pair present or not, indegree at most 2, acyclicity by source peeling) gives 1, 1, 3, 25, 443 for $0 \leq n \leq 4$, and a brute force over parent-set choices gives 13956 and 695902 for $n = 5, 6$. Independent code for the sink recurrence (2.2) reproduces these and equals the live b-file term by term for $0 \leq n \leq 100$ (the entry and the mirror file are still revision #9 of 12 July 2026); the columns $c = 2, \dots, 5$ of `data/counts.json` equal the recurrence with g_c ; the entry’s own Python program agrees with it for $n \leq 60$ (the write said $n \leq 40$); and $c = 1$ gives $(n + 1)^{n-1}$ for $n \leq 30$. The quotations, the author and date, the two-line proof (with $g(n - 1) = (n^2 - n + 2)/2$) and the congruences for $n \leq 100$ were confirmed, and the entry’s inequality holds for every $n \leq 100$. *Remark 6.1.* Every number of the table and of the fits was reproduced in 50-digit arithmetic; the logic (consistent under an unproved ansatz, no confirmation) is right. *Remark 7.1.* The instance of `p0:prop:factorial-core` was re-derived. The strict increase is proved: by (7.1), $a_n \geq na_{n-1}$ for $n \geq 2$ (the n sink extensions of a graph on $[n - 1]$ are distinct and admissible, and removing the sink recovers the graph), so $a_n \geq 2a_{n-1} > a_{n-1}$ for $n \geq 2$ and $N(x)$ is literally the staircase N_* with $n_0 = n_1 = 1$ (also checked numerically to $n = 100$). The classification of Theorem 1.2 as an analogue of `p0:thm:staircase(2)`, and the formal $m = 1$ term of the operator series, were confirmed. *Mallein.* Lemmas 2.6 and 2.7 of arXiv:1307.4496v3, with the hypotheses (2.12)–(2.13), were read again on the page images of pp. 11 and 13; the write’s note states exactly these, and agrees with the note of `a238873-diagonal-partitions` Part IV. Section 5 was re-read line by line: the upper bound uses only (2.15) at fixed h_- ; the lower bound uses only (2.19) at fixed h_+ with fixed scaled intervals (the containments (5.16) and (5.17) re-checked); the uniformity over block lengths is the elementary consequence of Remark 4.2; the initial segment and the terminal descent (probability exactly e^{-L_n}) are elementary; the identities (5.10) and (5.23) and the constants of Theorem 1.3 were re-derived; and no strip estimate, no time-inhomogeneous case and no exact endpoint is taken from Mallein. No mathematical error was found. One wording is corrected, with a dated note keeping the first wording: Remark 6.1 said that the unstable coefficient of $\log n$ “suggests no power prefactor”; that coefficient stays near $\frac{1}{2}$, and no power prefactor is inferred from it either way.

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